

- 2 Provide full-depth end diaphragms and full-depth intermediate diaphragms at quarter points along the span length.
- 3 Contact the SSDE for direction on non-standard steel floor beams or complex floor systems.

Modification for Non-Conventional Projects:

Delete Footnote 3 in Table 2.10-1 and replace with the following:

3 Do not use transverse floor beam systems for vehicular bridges unless permitted in the RFP.

Table 2.10-2 Redundancy Factors for Steel Box Girders

Number of Box Girders in Cross Section	η_R Factor ¹	
	With Exterior Diaphragms ²	Without Exterior Diaphragms
2	1.05	1.20

1 Top flange spacing more than 14-feet is not permitted.

2 With at least three evenly spaced exterior intermediate diaphragms (excluding end diaphragms) in each span. See **SDG** 5.6.3 for additional requirements.

The following requirements are applicable to the redundancy factor:

1. Applied at the component level. For girder components, the redundancy factor shall not be applied to the girder reactions for the bearing, substructure and foundation designs. For the substructure components including straddle or integral pier caps, the redundancy factor shall not be applied to the foundation if they are separate components (e.g. in a C-pier supported by a pile cap and piles, the redundancy factor is not applied to the pile design). For non-framed straddle or integral pier caps, the redundancy factor shall not be applied to the column and foundation designs.
2. Applied to flexural and axial effects of the component. For girder components, the redundancy factor is applied to splices, connections and cross-frames/ diaphragms or floor beam designs.
3. For steel components and mildly reinforced concrete components, apply the redundancy factor to the Strength I, II, and IV Limit State. For concrete post-tensioned components, apply the redundancy factor to the Strength I and Service I Limit State.
4. Applicable to Pedestrian bridges.
5. Not Applicable to **LRFD** 3.4.2, Load Factors for Construction Loads.

For special structures (e.g. cable, suspension, and other structures not normally used by FDOT), submit a redundancy factor to the SSDE for approval.